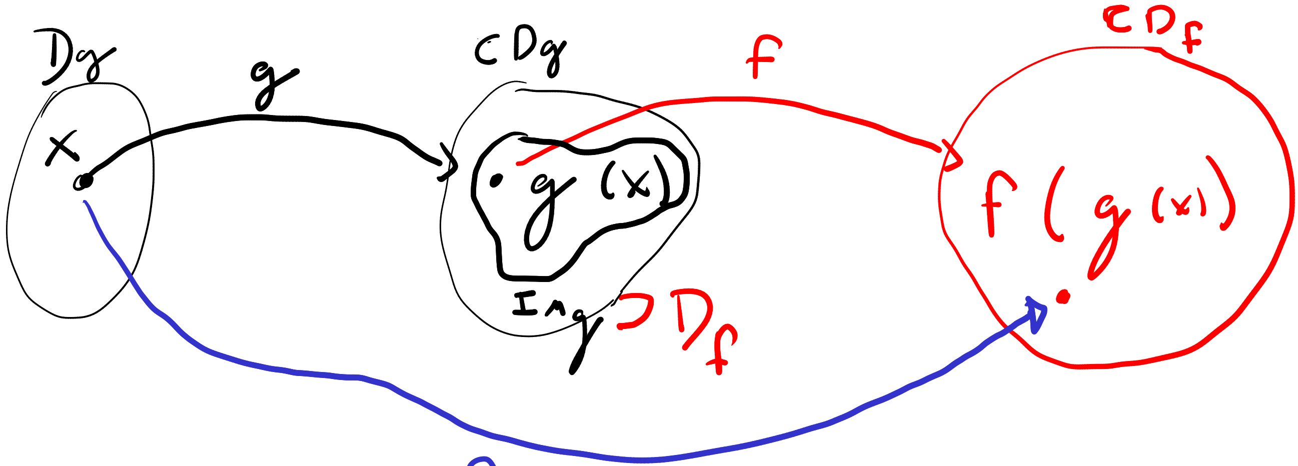
Regra da Cadeia ≡ função composta



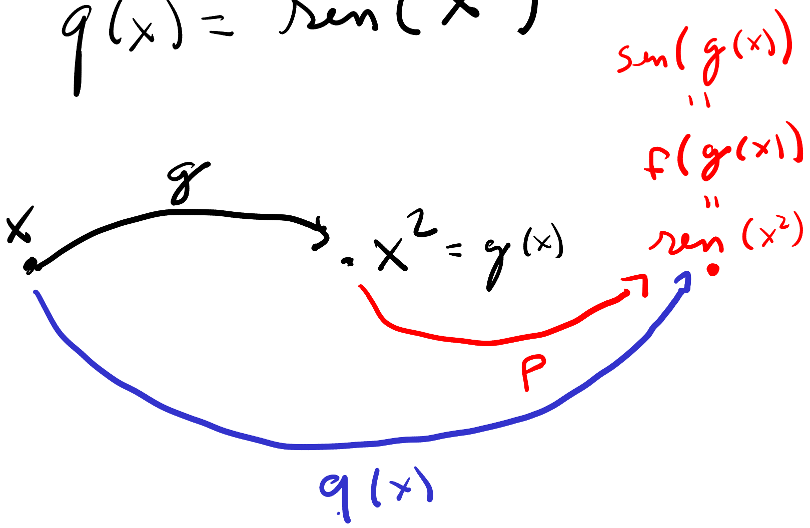
$$f \circ g(x) = f(g(x))$$

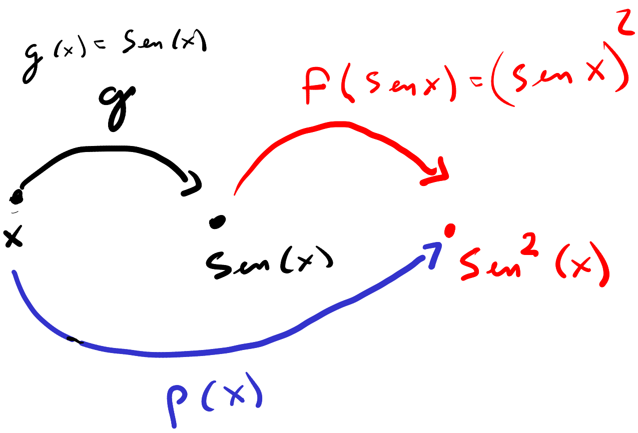
Ideia de função composta

Exemplos:

$$q(x) = \operatorname{sen}(x^2)$$

$$p(x) = \operatorname{sen}^2(x) = (\operatorname{sen}(x))^2$$





$q(x) = \operatorname{sen}(\underbrace{x^2}_{u})$

- $q = \operatorname{sen}(u)$
- $u = x^2$

$$\frac{dq}{dx} = \frac{dq}{du} \cdot \frac{du}{dx}$$

$$\frac{dq}{dx} = ?$$

Regra da Cadeia: $\frac{dq}{dx} = \boxed{\frac{dq}{du}}^{\cos(u)} \cdot \boxed{\frac{du}{dx}}^{2x} = \cos(u) \cdot 2x$

$= \cos(x^2) \cdot 2x$

$$p(x) = \text{sen}^2(x) = (\underbrace{\boxed{\text{sen}(x)}}_{u})^2 = u^2$$

- $u = \text{Sen}(x)$
- $p = u^2$

$$\frac{dp}{dx} = \underbrace{\frac{dp}{du}}_{2u} \cdot \underbrace{\frac{du}{dx}}_{\cos(x)}$$

$$\frac{dp}{dx} = 2u \cdot \cos(x)$$

$$\boxed{\frac{dp}{dx} = p'(x) = \underbrace{2\,\text{sen}(x)\cos(X)}_{\text{Sen}(2x)}}$$

Obs: $2\,\text{Sen}\,x\,\cos x = \text{sen}(2x)$

$\text{sen}(a \pm b) = \text{sen}\,a \cos b \pm \text{sen}\,b \cos a$ } $a = b = x$ $\text{Sen}(2x) = \text{sen}\,x\cos x + \text{sen}\,x \cos x = 2\,\text{sen}\,x \cos x$

$\cos(a \pm b) = \cos a \cos b \mp \text{sen}\,a \cos b$

$f(x) = e^{\overbrace{\boxed{\text{sen}(x)}}^{u}}$

$u = \text{Sen}\,x$

$F = e^u$

$\frac{df}{dx} = ?$

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx}$$

$$= e^u \cdot \cos x$$

$$= e^{\text{sen}\,x} \cdot \cos x$$

$$f(x) = e^{\overset{u}{\boxed{(\text{sen}\,x)^2}}} \qquad \frac{df}{dx} = ?$$

- $u = (\text{sen}\,x)^2$
- $f = e^u$

$$\frac{df}{dx} = \frac{df}{du}\cdot\frac{du}{dx}$$

$$= e^u \cdot \underbrace{\frac{d\big(\overset{h(x)}{(\underbrace{\text{sen}\,x}_{v})^2}\big)}{dx}}$$

$v = \text{sen}\,x$

$(\text{sen}\,x)^2 = v^2 = h$

$$\frac{d(\text{sen}\,x)^2}{dx} = \frac{d(\text{sen}\,x)^2}{dv}\cdot\frac{dv}{dx}$$

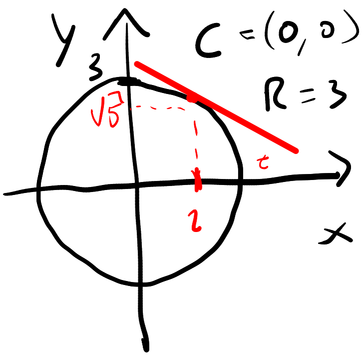
$$= 2v\cdot\cos x$$

$$= 2\,\text{sen}\,x\,\cos x$$

$$= e^{(\text{sen}\,x)^2}\cdot 2\,\text{sen}\,x\,\cos x$$

Derivação Implícita ≡ Aplicação da Regra da Cadeia, quando for difícil (ou impossível) explicitar f(x).

Exemplo: achar o coeficiente angular da reta tangente à circunferência no ponto $(2, \sqrt{5})$



$$(x-0)^2 + (y-0)^2 = 3^2$$

$$x^2 + y^2 = 9$$

Função implícita: $y = f(x)$

$$\frac{d}{dx}\left(x^2 + (f(x))^2\right) = \frac{d}{dx}(9)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}\left(\underbrace{f(x)}_{\text{função composta}}\right)^2 = 0$$

$u = f(x)$

$$\frac{d\,u^2}{dx} = \frac{d\,u^2}{du} \cdot \frac{du}{dx} = 2u \cdot \frac{df(x)}{dx}$$

$$\underbrace{2x + 2f(x) \cdot \frac{df}{dx}}_{\text{REGRA DA CADEIA}} = 0$$

$$\frac{df}{dx} = -\frac{2x}{2f(x)} = \boxed{-\frac{x}{f(x)} = \frac{df}{dx}(x)}$$

Mas $m_t = \dfrac{df}{dx}(2) = -\dfrac{2}{f(2)} = -\dfrac{2}{\sqrt{5}}$

LISTA DE EXERCÍCIOS (LIVRO CAP. 4) Seção 4.7:

4.19; 4.20; 4.23; 4.24;
4.26; 4.27; 4.28; 4.29; 4.30;
4.32; 4.33; 4.35; 4.36;
4.37; 4.40; 4.41; 4.43; 4.44

Seção 5.7 (p 176) → 5.21, 5.22, 5.23; 5.24; 5.25; 5.26;
5.27; 5.28